Beetle video review

What we want to learn from the dyadic and six-beetle videos

The goal

Help us decide whether the tracking is reliable enough to extract useful behavioral measurements, where the code still needs improvement, and which quality-control checks would identify data that need closer review. We expect some mistakes; the key question is whether they change the conclusions we would draw.

2026 dyadic interactions	2024 six-beetle interactions
25 videos in the documented review set. Vince's initial impression: tracking looks quite good.	26 videos in the documented review set. Vince's initial impression: quality varies substantially.

Counts follow the October 5 project handoff. These are review impressions, not established accuracy results; all automatic outputs remain unreviewed.

The measurements we hope to extract

Time near the wall: within one body length of the arena wall.
Time on the fungus: whether the beetle is correctly located on the fungus.
Number of interactions: whether encounters between individuals are detected and separated correctly.

What to watch for

Review question	What counts as useful evidence?
1. Are the locations correct?	Does each plotted position match the visible beetle? Note missing positions, jumps, and errors near the wall or fungus.
2. Are identities maintained?	Follow individuals into and out of encounters. Look for identity swaps, especially after overlap or reappearance.
3. Is PSC believable?	PSC means presumed social contact. Compare inferred positions with the visible animals. Does the output recognize separation and recover the correct identities and positions?
4. Is first movement captured?	When a beetle first gets up and walks away, does tracking capture the start, the correct animal, and its path? Note delayed detection, missing movement, or apparent movement caused by a tracking jump.

Compare the original video with the tracking and PSC display. A presumed location is an inference, not a directly observed position. Record uncertainty when the video cannot resolve it.

How to record your review

A consistent spreadsheet will help us distinguish isolated mistakes from recurring problems.

Use two spreadsheet tabs

Tab	Suggested columns
Video summary One row per video reviewed	Video/run name; dyadic or six-beetle; reviewer and date; exact portion watched; full or partial review; confidence in wall time, fungus time, interaction count, and first movement; main concerns; recommended follow-up.
Error log One row per error episode	Video/run name; beetle ID(s); start/end frame or timestamp; error type; what the video shows; what the output shows; affected measurement(s); impact; confidence or uncertainty; screenshot/reference if helpful.

Always identify the frame system. Say whether a number is a source/global frame, review-video frame, or playback timestamp. Record a range for an ongoing error, rather than treating every affected frame as a separate incident.

Suggested error types and impact notes

Use consistent labels such as **wrong location, identity swap, missing tracking, implausible PSC, separation missed, incorrect identity recovery,** or **first movement missed/delayed**. Add another label if needed.

Describe whether an error is brief and unlikely to affect a measurement, changes a specific measurement, or makes a segment difficult to use. Mark **uncertain** when appropriate. A video may be useful for one measurement and unreliable for another.

Which quality-control checks would help?

Look for patterns linking visible errors to potential warning metrics:

Potential warning	Question to answer during review
Missing frames or long gaps	Do gaps hide encounters or the first movement? Is their duration more informative than their total number?
Large position jumps or identity instability	Do warnings coincide with real errors? Does the apparent jump reflect a camera shift rather than animal movement?
Unusual interaction counts or long PSC periods	Do these indicate missed separations or repeated counting of one encounter, or are they real behavior?
Turtling / on-back episodes	Does this posture coincide with tracking problems? Note visible examples; do not assume an automatic flag proves posture.

What your review should help us decide

Which recurring errors need a code change? Which remaining mistakes are acceptable for each intended measurement? Which warnings reliably identify problematic segments or videos? Recommend cases for exclusion review, with reasons; do not treat a warning alone as an exclusion rule.

Before calculating final measurements, we will need shared definitions for body length, the fungus boundary, and what counts as a new interaction. For now, flag borderline examples. Unseen footage and uncertain observations must not be recorded as error-free.

Prepared by Codex from Vince's notes, October 9, 2026. Review-set counts and PSC status follow the supplied project handoff.